



TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC2 2021

H2 BIOLOGY

Paper 4 Practical

CONFIDENTIAL INSTRUCTIONS

9744/04

30 August 2021

2 hours 30 minutes

This document gives details of how to prepare for and administer the practical exam.

Great care should be taken to ensure that any confidential information, including the identity of material on microscope slides where appropriate, does not reach the candidates directly or indirectly.

This document consists of **5** printed pages and **3** blank pages.



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Question 1

Preparation List

	Apparatus/Reagents/Chemicals	Quantity per candidate
1	Agar containing sodium hydroxide and bromothymol blue solution on a white tile, labelled B , covered by a damp paper towel.	1 block
2	0.02 mol dm ⁻³ hydrochloric acid in a beaker or container, labelled A provided at room temperature.	At least 100 cm ³
3	0.01 mol dm ⁻³ hydrochloric acid in a beaker or container, labelled X provided at room temperature.	At least 60 cm ³
4	Distilled / tap water, in a suitable container labelled W (also used in Question 2)	at least 100 cm ³
5	10 cm ³ syringes	2
6	Beakers or containers (minimum capacity approximately 50 cm ³)	6
7	Blunt scalpel (also used in Question 2)	1
8	White tile	1
9	Forceps	1
10	Ruler, marked in mm (also used in Question 2)	
11	Glass rod	1
12	(About 200 cm ³) container labelled For washing	1
13	(About 200 cm ³) Container, labelled Waste (also used in Question 2)	1
14	Paper towels	4
15	Glass marker pen, permanent (also used in Question 2)	1
16	Stopwatch	1
17	Suitable eye protection (safety goggles)	1

Instructions for preparation for Question 1

The agar containing sodium hydroxide and bromothymol blue solution, labelled **B**, may be prepared the day before the examination.

To prevent drying out, the containers may be wrapped in a plastic bag and placed in a fridge overnight. The agar may be removed to reach room temperature before the start of the examination.

Agar containing bromothymol blue indicator labelled **B**

Dissolve bromothymol blue indicator using sodium hydroxide solution.

To prepare **0.01 mol dm⁻³ sodium hydroxide solution** for the bromothymol blue indicator:

- Add 4g sodium hydroxide with 80cm³ distilled water in a beaker.
- Mix well to dissolve.
- Make up 100cm³ with distilled water. This is the stock solution of 1 mol dm⁻³ sodium hydroxide solution.
- Put 1cm³ of this stock solution into a beaker or container and make up to 100cm³ with distilled water.
- This makes 0.01 mol dm⁻³ sodium hydroxide solution.

To prepare **bromothymol blue indicator**:

- Put 0.1g of bromothymol blue powder into a beaker or container.
- Put 16cm³ of 0.01 mol dm⁻³ sodium hydroxide solution into the same beaker.
- Mix well to dissolve.
- Make up to 250cm³ with distilled water.

To prepare **agar**:

- Put 10g of agar ('technical' not 'nutrient') into 450cm³ of distilled water.
- Heat to 95°C, stirring until dissolved.
- Allow to cool to 60°C.
- Put 50cm³ of bromothymol blue indicator into the cooled agar and mix well.
- If the colour of the agar is green, add additional 0.01 mol dm⁻³ sodium hydroxide solution to the agar until a blue colour is achieved.
- Put the agar into shallow, flat containers so that the depth of the agar is between 0.5 cm and 0.7 cm.
- Place the containers on a horizontal flat surface so that the agar sets to an even depth. Keep the agar in cool conditions in order to set firmly.

Note: It is essential that the agar **does not dry out** after it has set. To prevent drying out, the containers may be wrapped in a plastic bag or similar cover.

Just before the examination, cut the agar into blocks measuring at least 6cm x 4cm as shown in Fig. 1.1., page 6 of question paper.

Cut enough blocks so that each candidate has one block of at least 6cm x 4cm, on a white tile, covered by a damp paper towel.



Question 2

Preparation List

Each candidate will have sole, uninterrupted use of a microscope for 1 hour 15 minutes only.

For each candidate:

- the microscope must be set up on the low-power objective lens
- the slide must **not** be left on the stage of the microscope

Each candidate will require:

	Apparatus/Reagents/Chemicals	Quantity per candidate
1	Microscope with: • an eyepiece lens, with x10 magnification • a low-power objective lens, x10 magnification • a high-power objective lens, x40 magnification	1 between 2
2	A piece of peeled potato (e.g. Irish) wrapped in a damp paper towel, in a base of a Petri dish labelled T	1
3	Leaf, in a beaker labelled L	1
4	Mortar and pestle	1 set
5	Iodine (in potassium iodide) solution, in a suitable container labelled iodine solution and provided with a dropper	at least 30 cm ³
6	Distilled water, in a suitable container labelled W (also used in Question 1)	at least 10 cm ³
7	Microscope slide and coverslip	6
8	Mounted needle	1
9	Pasteur pipette	1
10	Glass rod	1
11	Glass marker pen, permanent	1
12	Ruler, marked in mm (also used in Question 1)	1
13	Scalpel, sharp (also used in Question 1)	1
14	Black card, approximately 10cm x 10cm	1
15	Paper towels	4
16	Container, labelled waste (also used in Question 1)	1

Instructions for preparation for Question 2

A piece of peeled potato (e.g. Irish), in a base of a Petri dish labelled T

Use a cork borer with a diameter of about 13mm (e.g. a number 7 cork borer) to obtain cylinders of tissue from potato tubers (e.g. Irish potato).

Cylinders should be wrapped in a damp paper towel. The exact dimensions are not important (e.g. approximately 5 cm × 2 cm × 2 cm).

A fresh potato piece should be provided for each candidate.

Leaf suspension, in a base of a Petri dish labelled L

Cut up about 10cm² of lettuce leaf into pieces, then made into a suspension with no more than 3cm³ of water using either a blender or a pestle and mortar.

It is important that the suspension of the lettuce cells is fresh and that the Supervisor checks that the suspension of lettuce leaf is fine enough to show the epidermal cells, guard cells, mesophyll cells and xylem vessels which should be visible under the microscope. (If green lettuce is not available then cabbage or Chinese lettuce leaves may also be used).

Iodine (in potassium iodide) solution, in a suitable container labelled I

Standard iodine solution suitable for testing food for starch can be used, e.g. commercially prepared 0.1N/0.05mol dm⁻³ iodine solution.

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